Math 257 Definitions and Theorems

Week 13: Modules 41-43

Fall 2023

41 Singular Value Decomposition

Definition 41.1. Let A be an $m \times n$ matrix. A **singular value decomposition** of A is a decomposition $A = U\Sigma V^T$ where

- U is an $m \times m$ matrix with orthonormal columns,
- Σ is an $m \times n$ rectangular diagonal matrix with non-negative numbers on the diagonal,
- V is an $n \times n$ matrix with orthonormal columns.

Remark. The diagonal entries $\sigma_i = \Sigma_{ii}$ which are positive are called the **singular values** of A. We usually arrange them in decreasing order, that is $\sigma_1 \geq \sigma_2 \geq \dots$

Remark. Let A be an $m \times n$ matrix with rank r. Recall that

- $\text{Nul}(A^TA) = \text{Nul}(A)$ and $\text{Nul}(AA^T) = \text{Nul}(A^T)$.
- A^TA is symmetric and has rank r.

Algorithm 41.1. Let A be an $m \times n$ matrix with rank r.

- Find orthonormal eigenbasis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ of A^TA with eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > \lambda_{r+1} = 0 = \dots = \lambda_n$.
- Set $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, \dots, n$.
- Set $\mathbf{u}_1 = \frac{1}{\sigma_1}A\mathbf{v}_1$, …, $\mathbf{u}_r = \frac{1}{\sigma_r}A\mathbf{v}_r$. (**Magic**: orthonormal!)
- Find $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m \in \mathbb{R}^m$ such that $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ is an orthonormal basis of $\mathbb{R}^m$.
- Set

$$U = \begin{bmatrix} \mathbf{u}_1 & \dots & \mathbf{u}_m \end{bmatrix}, \Sigma = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_{\min\{m,n\}} \end{bmatrix}, V = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_n \end{bmatrix}$$

Notice that $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n \in \text{Nul}(A^TA)$, and hence $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n \in \text{Nul}(A)$.

$$\begin{aligned} \rightsquigarrow AV &= A\begin{bmatrix}\mathbf{v}_1 & \ldots & \mathbf{v}_n\end{bmatrix} = \begin{bmatrix}A\mathbf{v}_1 & \ldots & A\mathbf{v}_n\end{bmatrix} \\ &= \begin{bmatrix}\sigma_1\mathbf{u}_1 & \ldots & \sigma_r\mathbf{u}_r & \mathbf{0} & \ldots & \mathbf{0}\end{bmatrix} = U\Sigma \rightsquigarrow A = U\Sigma V^T. \end{aligned}$$

Theorem 41.1. *Let A be an $m \times n$ matrix with rank r, and suppose $U = \begin{bmatrix}\mathbf{u}_1 & \ldots \mathbf{u}_m\end{bmatrix}$, $V = \begin{bmatrix}\mathbf{v}_1 & \ldots \mathbf{v}_n\end{bmatrix}$, and Σ are such that $A = U\Sigma V^T$ is a SVD of A. Then*

- $(\mathbf{u}_1, \ldots, \mathbf{u}_r)$ *is a basis of* $\text{Col}(A)$.
- $(\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m)$ *is a basis of* $\text{Nul}(A^T)$.
- $(\mathbf{v}_1, \ldots, \mathbf{v}_r)$ *is a basis of* $\text{Col}(A^T)$.
- $(\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n)$ *is a basis of* $\text{Nul}(A)$.

42 Low Rank Approximation

Theorem 42.1. *Let A be an $m \times n$ matrix with rank r, and let $U = \begin{bmatrix}\mathbf{u}_1 & \ldots \mathbf{u}_m\end{bmatrix}$, $V = \begin{bmatrix}\mathbf{v}_1 & \ldots \mathbf{v}_n\end{bmatrix}$ be matrices with with orthonormal columns and Σ be a rectangular diagonal $m \times n$ matrix such that $A = U\Sigma V^T$ is an SVD of A. Then*

$$\begin{aligned} A &= \sigma_1\mathbf{u}_1\mathbf{v}_1^T + \sigma_2\mathbf{u}_2\mathbf{v}_2^T + \cdots + \sigma_r\mathbf{u}_r\mathbf{v}_r^T \\ &= \begin{bmatrix}\mathbf{u}_1 & \ldots & \mathbf{u}_r\end{bmatrix} \begin{bmatrix}\sigma_1 & & \\ & \ddots & \\ & & \sigma_r\end{bmatrix} \begin{bmatrix}\mathbf{v}_1 & \ldots & \mathbf{v}_r\end{bmatrix}^T. \end{aligned}$$

Definition 42.1. For $k \leq r$, define

$$A_k = \sigma_1\mathbf{u}_1\mathbf{v}_1^T + \sigma_2\mathbf{u}_2\mathbf{v}_2^T + \cdots + \sigma_k\mathbf{u}_k\mathbf{v}_k^T.$$

Idea. If $\sigma_1 \gg \sigma_2 \gg \ldots ...$, then A_k is a good approximation of A.

Definition 42.2. Let A be an $m \times n$ matrix with rank r. A **compact singular value decomposition** of A is a decomposition $A = U_c\Sigma_c V_c^T$ where

- $U_c = \begin{bmatrix}\mathbf{u}_1 & \ldots \mathbf{u}_r\end{bmatrix}$ is an $m \times r$ matrix with orthonormal columns,
- Σ_c is an $r \times r$ diagonal matrix with positive diagonal elements,
- $V_c = \begin{bmatrix}\mathbf{v}_1 & \ldots \mathbf{v}_r\end{bmatrix}$ is an $n \times r$ matrix with orthonormal columns.

43 Pseudoinverses

Definition 43.1. Let A be an $m \times n$ matrix with rank r. Given the compact singular value decomposition $A = U_c\Sigma_c V_c^T$, where

- $U_c = \begin{bmatrix} \mathbf{u}_1 & \dots \mathbf{u}_r \end{bmatrix}$ is an $m \times r$ matrix with orthonormal columns,
- Σ_c is an $r \times r$ diagonal matrix with positive diagonal elements,
- $V_c = \begin{bmatrix} \mathbf{v}_1 & \dots \mathbf{v}_r \end{bmatrix}$ is an $n \times r$ matrix with orthonormal columns,

we define the **pseudoinverse** A^+ of A as $V_c \Sigma_c^{-1} U_c^T$.

Theorem 43.1. *Let* $\mathbf{v} \in \mathrm{Col}(A^T)$ *and* $\mathbf{w} \in \mathrm{Col}(A)$. *Then* $A^+ A\mathbf{v} = \mathbf{v}$ *and* $AA^+\mathbf{w} = \mathbf{w}$.

Remark. If A is $n \times n$ and invertible, then $\mathrm{Col}(A) = \mathbb{R}^n$. Thus $A^{-1} = A^+$.

Theorem 43.2. *Let* A *be an* $m \times n$ *matrix and let* $\mathbf{b} \in \mathbb{R}^m$. *Then* $A^+\mathbf{b}$ *is the LSQ solution of* $A\mathbf{x} = \mathbf{b}$ *(with minimum length).*

Remark. This is particularly useful, when solving many different LSQ problems of the form $A\mathbf{x} = \mathbf{b}$, where A stays the same, but $\mathbf{b}$ varies.